ROOT CAUSE ANALYSIS (RCA)

WhatsApp CTA Link Failure on Selected Android Devices

Prepared For: TalentWale

Subject: WhatsApp In-App Browser Link Opening Issue

Executive Summary

Users reported that a WhatsApp CTA link opened successfully on some devices while failing on others. The issue was reproduced and investigated using Android Logcat analysis, device comparison, Chrome validation, and SSL configuration assessment. The investigation identified an SSL certificate chain configuration issue causing TLS handshake failures within WhatsApp's embedded browser environment.

Incident Description

A CTA URL distributed through WhatsApp failed to open for a subset of Android users. The same URL was accessible through Chrome, creating inconsistent user behavior across devices.

Investigation Timeline

Step	Activity
1	Collected Logcat from failing device
2	Collected Logcat from working device
3	Compared WhatsApp IAB behavior
4	Validated Android, Chrome and WebView versions
5	Executed direct Chrome launch tests
6	Performed SSL Labs assessment on talentwale.com

Evidence Collected

Failing Device: Chromium reported SSL handshake failures with net_error -202 during page load in WhatsApp In-App Browser.

Working Device: WhatsApp In-App Browser launched successfully without SSL handshake errors.

Chrome Validation: The same URL opened successfully in Chrome on the affected device.

SSL Labs Result: SSL grade B with warning: 'This server's certificate chain is incomplete.'

Technical Analysis

The repeated TLS handshake failures observed in WhatsApp In-App Browser indicate that certificate validation fails before page rendering. The behavior differs from Chrome, suggesting that WhatsApp's embedded browser/WebView is unable to build a trusted certificate path for the target domain under certain Android environments.

Root Cause

Primary Root Cause: Incomplete SSL certificate chain configuration on talentwale.com. The server is not consistently providing the full certificate chain (including required intermediate certificates), causing TLS validation failures in WhatsApp In-App Browser / Android WebView on specific Android versions.

Business Impact

Users attempting to apply through WhatsApp CTA links may be unable to access the landing page, leading to reduced application conversions and inconsistent user experience across devices.

Recommended Corrective Actions

1. Install and serve the complete SSL certificate chain (fullchain certificate).
2. Verify intermediate certificates are correctly configured.
3. Re-test using SSL Labs until Grade A/A+ is achieved.
4. Validate functionality on Android WebView and WhatsApp In-App Browser after deployment.
5. Monitor CTA conversion metrics after SSL remediation.

Conclusion

Based on log analysis, browser comparison testing, and SSL Labs validation, the issue is attributed to SSL certificate chain configuration. Confidence level: 95–99%.